

# Method: Shared-Backbone Step-wise PPO with Episode-Level Relative Correction

## Method

We consider long-horizon agentic reinforcement learning where an episode consists of multiple environment interaction steps. At each step, the agent observes a context, generates an action such as a tool call, and receives an environment reward. Our goal is to construct a step-wise actor-critic algorithm while preserving the episode-level relative training signal used by GRPO/GiGPO-style grouped rollouts. The key design choice is to avoid maintaining a separate critic model. Instead, we attach a value head to the actor model and use the shared actor backbone to predict state values.

### Problem formulation

Let  $x$  denote an initial task instance sampled from a training distribution. A rollout trajectory generated by policy  $\pi_\theta$  is

$$\tau = (s_0, a_0, r_0, s_1, a_1, r_1, \dots, s_{T-1}, a_{T-1}, r_{T-1}),$$

where  $s_t$  is the agent context at environment step  $t$ ,  $a_t$  is the complete action generated by the language model, and  $r_t \in \mathbb{R}$  is the immediate environment reward. Although  $a_t$  is represented by a sequence of response tokens,

$$a_t = (y_{t,1}, y_{t,2}, \dots, y_{t,L_t}),$$

we treat the whole response as a single environment-level action for value estimation, return computation, and advantage estimation.

For grouped rollouts, each initial task instance  $x$  is repeated  $K$  times. The resulting trajectories share the same episode-group identifier  $g = \text{uid}(x)$ , while each trajectory has a unique trajectory identifier  $m = \text{traj}(\tau)$ . We denote the set of trajectories in group  $g$  by  $\mathcal{T}_g$ .

### Shared actor-value model

The actor is a causal language model with parameters  $\theta$ . Instead of creating a standalone critic backbone, we attach a regression head  $v_\psi$  to the actor backbone. Given the hidden states of the actor model,

$$H_t = f_\theta(s_t, a_t) \in \mathbb{R}^{n_t \times d},$$

where  $n_t$  is the full context-plus-response sequence length, we define the strict state-value estimate using the hidden state at the action-generation boundary:

$$V_{\theta,\psi}(s_t) = v_\psi(H_t[b_t]), \quad b_t = n_t - L_t - 1.$$

Here  $b_t$  indexes the final context token before the first generated action token. Because the actor is causal,  $H_t[b_t]$  does not attend to future response tokens. Therefore,  $V_{\theta,\psi}(s_t)$  is a state-value estimate rather than an action-conditioned value estimate.

The value head is implemented as a small MLP,

$$v_\psi(h) = W_2 \sigma(W_1 h + b_1) + b_2,$$

where  $h \in \mathbb{R}^d$ ,  $W_1 \in \mathbb{R}^{d_h \times d}$ ,  $W_2 \in \mathbb{R}^{1 \times d_h}$ , and  $\sigma$  is GELU. A linear head  $v_\psi(h) = w^\top h + b$  is a valid special case. In the main method, gradients from the value loss are allowed to update both  $\psi$  and the shared backbone  $\theta$ , with a small value-loss coefficient. Detaching the backbone for the value loss is reserved as an ablation.

**Action-conditioned variants.** One may instead pool hidden states over response tokens or use the last response token value. This yields an action-conditioned quantity  $V(s_t, a_t)$  or  $Q(s_t, a_t)$  rather than a strict state-value baseline. We do not use this in the main method, but retain it as an ablation.

## Step-wise returns and GAE

For each trajectory  $\tau$ , we compute step-wise temporal-difference residuals along the environment-step axis. Let  $d_t \in \{0, 1\}$  indicate whether step  $t$  is terminal. With discount  $\gamma \in [0, 1]$  and GAE parameter  $\lambda \in [0, 1]$ , define

$$\delta_t = r_t + \gamma(1 - d_t)V_{\theta, \psi}^{\text{old}}(s_{t+1}) - V_{\theta, \psi}^{\text{old}}(s_t),$$

where the old values are evaluated before policy optimization on the collected batch. The step-wise GAE is computed backward over environment steps:

$$A_t^{\text{step}} = \delta_t + \gamma\lambda(1 - d_t)A_{t+1}^{\text{step}}, \quad A_T^{\text{step}} = 0.$$

The corresponding value target is

$$R_t^{\text{step}} = A_t^{\text{step}} + V_{\theta, \psi}^{\text{old}}(s_t).$$

Equivalently, when  $\lambda = 1$ ,  $R_t^{\text{step}}$  reduces to the Monte Carlo discounted return

$$G_t = \sum_{k=t}^{T-1} \gamma^{k-t} r_k.$$

All quantities above are indexed by environment step, not by response-token position.

## Episode-level relative advantage

Let the undiscounted episode return of trajectory  $\tau_i \in \mathcal{T}_g$  be

$$R_i^{\text{epi}} = \sum_{t=0}^{T_i-1} r_{i,t}.$$

Following GRPO-style group-relative normalization, the episode-level advantage is

$$A_i^{\text{epi}} = \frac{R_i^{\text{epi}} - \mu_g}{\sigma_g + \epsilon}, \quad \mu_g = \frac{1}{|\mathcal{T}_g|} \sum_{\tau_j \in \mathcal{T}_g} R_j^{\text{epi}},$$

$$\sigma_g^2 = \frac{1}{|\mathcal{T}_g| - 1} \sum_{\tau_j \in \mathcal{T}_g} (R_j^{\text{epi}} - \mu_g)^2.$$

If  $|\mathcal{T}_g| = 1$ , we set  $\mu_g = 0$  and  $\sigma_g = 1$  for numerical stability. This trajectory-level scalar is broadcast to every step in the trajectory:

$$A_{i,t}^{\text{epi}} = A_i^{\text{epi}}, \quad t = 0, 1, \dots, T_i - 1.$$

## Weighted correction of episode advantages

The final advantage combines the episode-level relative signal and the step-wise actor-critic signal. Before mixing, both advantages are normalized over valid steps in the batch:

$$\hat{A}_t^{\text{epi}} = \text{Norm}(A_t^{\text{epi}}), \quad \hat{A}_t^{\text{step}} = \text{Norm}(A_t^{\text{step}}).$$

We use a GiGPO-style weighted correction:

$$A_t^{\text{final}} = \frac{\hat{A}_t^{\text{epi}} + w_{\text{step}} \hat{A}_t^{\text{step}}}{1 + w_{\text{step}}},$$

where  $w_{\text{step}} \geq 0$  controls the strength of the step-wise critic correction. Setting  $w_{\text{step}} = 0$  recovers pure episode-level GRPO-style training, while larger values place more weight on step-wise credit assignment.

## Policy objective

Let  $\pi_{\theta_{\text{old}}}$  be the behavior policy used to sample the current batch. For each action token  $y_{t,\ell}$ , define the token-level importance ratio

$$\rho_{t,\ell}(\theta) = \exp(\log \pi_{\theta}(y_{t,\ell} \mid s_t, y_{t,<\ell}) - \log \pi_{\theta_{\text{old}}}(y_{t,\ell} \mid s_t, y_{t,<\ell})).$$

In the first implementation, the scalar step advantage  $A_t^{\text{final}}$  is broadcast to all valid response tokens in action  $a_t$ . The clipped PPO policy loss is

$$\mathcal{L}_{\text{pg}}(\theta) = -\mathbb{E}_{t,\ell} [\min(\rho_{t,\ell}(\theta) A_t^{\text{final}}, \text{clip}(\rho_{t,\ell}(\theta), 1 - \varepsilon, 1 + \varepsilon) A_t^{\text{final}})],$$

where the expectation is taken over valid action tokens. To reduce length bias, the main configuration uses sequence-level averaging over action tokens. A stricter future variant will replace token-level ratios with action-level ratios,

$$\rho_t^{\text{act}}(\theta) = \exp\left(\sum_{\ell=1}^{L_t} \log \pi_{\theta}(y_{t,\ell} \mid s_t, y_{t,<\ell}) - \log \pi_{\theta_{\text{old}}}(y_{t,\ell} \mid s_t, y_{t,<\ell})\right),$$

or a length-normalized version of the same quantity.

## Value objective

The shared value head is trained with a clipped scalar value loss. Let

$$V_t = V_{\theta,\psi}(s_t), \quad V_t^{\text{old}} = V_{\theta_{\text{old}},\psi_{\text{old}}}(s_t).$$

The clipped value prediction is

$$\tilde{V}_t = V_t^{\text{old}} + \text{clip}(V_t - V_t^{\text{old}}, -\varepsilon_v, \varepsilon_v).$$

The value loss is

$$\mathcal{L}_v(\theta, \psi) = \mathbb{E}_t \left[ \max\left((V_t - R_t^{\text{step}})^2, (\tilde{V}_t - R_t^{\text{step}})^2\right) \right].$$

This loss is scalar per environment step, even though the underlying language model can produce hidden states at every token position.

## KL regularization and total loss

We use a reference policy  $\pi_{\text{ref}}$  and add a token-level KL penalty to the actor loss. Let

$$\mathcal{L}_{\text{KL}}(\theta) = \mathbb{E}_{t,\ell} [D_{\text{KL}}(\pi_{\theta}(\cdot \mid s_t, y_{t,<\ell}) \parallel \pi_{\text{ref}}(\cdot \mid s_t, y_{t,<\ell}))].$$

The final optimization objective is

$$\mathcal{L}(\theta, \psi) = \mathcal{L}_{\text{pg}}(\theta) + c_v \mathcal{L}_v(\theta, \psi) + \beta \mathcal{L}_{\text{KL}}(\theta) - c_H \mathcal{H}(\pi_{\theta}),$$

where  $c_v$  is the value-loss coefficient,  $\beta$  is the KL coefficient, and  $c_H$  is the entropy coefficient.

## Implementation notes

The value head is attached to the actor module before FSDP wrapping, so it is optimized and check-pointed together with the actor. Since inference engines such as vLLM or SGLang only consume language-model weights, parameters whose names contain `value_head` must be filtered out before synchronizing actor weights to rollout engines. The rollout data must also store an explicit step index, because later batch reordering can otherwise destroy the environment-step order required by step-wise GAE.

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**Algorithm 1** Shared-Backbone Step-wise PPO with Episode-Level Relative Correction

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**Input:** Actor policy  $\pi_\theta$  with value head  $v_\psi$ , reference policy  $\pi_{\text{ref}}$ , grouped environments, group size  $K$ , discount  $\gamma$ , GAE parameter  $\lambda$ , step weight  $w_{\text{step}}$

**Output:** Updated actor and value-head parameters  $(\theta, \psi)$

- 1: **for** each training iteration **do**
  - 2:   Sample a batch of task instances and create  $K$  grouped rollouts for each task
  - 3:   Collect step samples  $(s_t, a_t, r_t, \text{uid}, \text{traj}, \text{step})$  from the agent-environment loop
  - 4:   Recompute old token log-probabilities under  $\pi_{\theta_{\text{old}}}$
  - 5:   Compute old state values  $V_t^{\text{old}} = v_{\psi_{\text{old}}}(H_t[b_t])$
  - 6:   Compute episode returns  $R_i^{\text{epi}}$  for each trajectory
  - 7:   Compute group-relative episode advantages  $A_i^{\text{epi}}$  and broadcast them to steps
  - 8:   **for** each trajectory  $\tau_i$  **do**
  - 9:     Sort its samples by step
  - 10:    Compute  $A_t^{\text{step}}$  and  $R_t^{\text{step}}$  by backward step-wise GAE
  - 11:    Normalize  $A^{\text{epi}}$  and  $A^{\text{step}}$  over valid steps
  - 12:    Compute  $A_t^{\text{final}} = (A_t^{\text{epi}} + w_{\text{step}} A_t^{\text{step}}) / (1 + w_{\text{step}})$
  - 13:    Broadcast  $A_t^{\text{final}}$  to valid response tokens for the PPO actor loss
  - 14:   **for** each PPO epoch and mini-batch **do**
  - 15:     Recompute current token log-probabilities under  $\pi_\theta$
  - 16:     Recompute current state values  $V_t = v_\psi(H_t[b_t])$
  - 17:     Compute clipped policy loss  $\mathcal{L}_{\text{pg}}$
  - 18:     Compute clipped scalar value loss  $\mathcal{L}_{\text{v}}$  using  $R_t^{\text{step}}$
  - 19:     Compute KL loss against  $\pi_{\text{ref}}$
  - 20:     Update  $(\theta, \psi)$  with  $\mathcal{L} = \mathcal{L}_{\text{pg}} + c_v \mathcal{L}_{\text{v}} + \beta \mathcal{L}_{\text{KL}} - c_H \mathcal{H}$
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